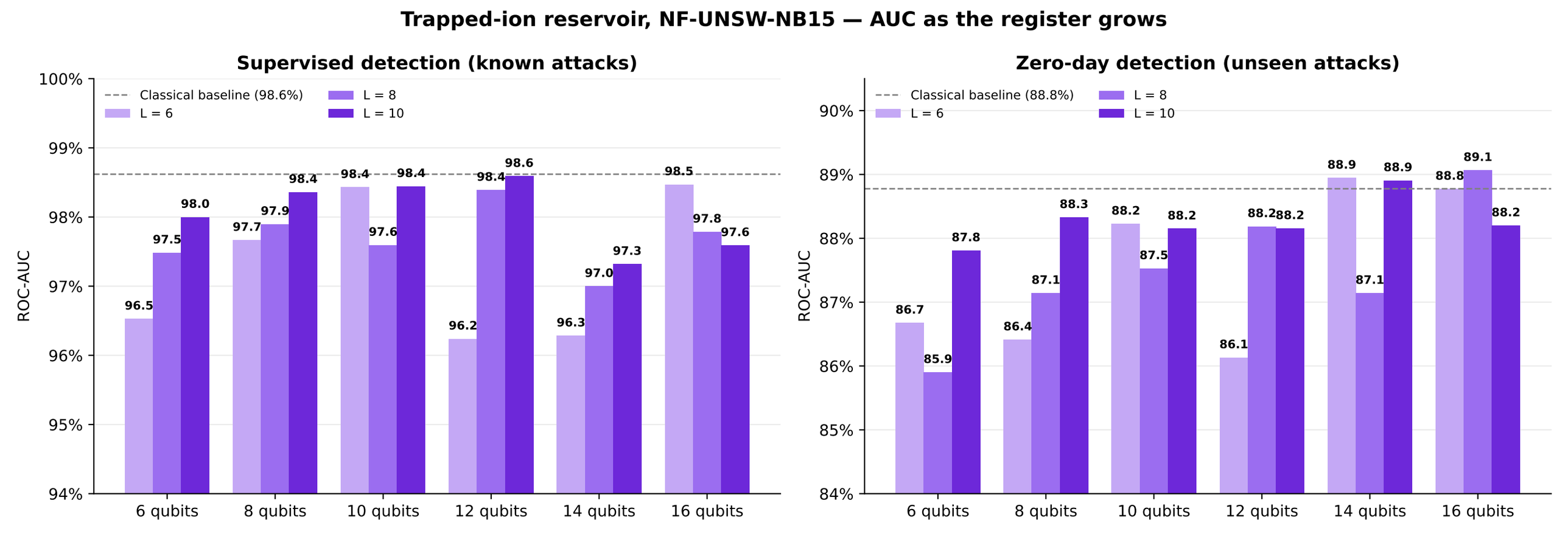
Trapped-ion reservoir, NF-UNSW-NB15 — AUC as the register grows
Supervised detection (known attacks)
Classical baseline (98.6%)
L = 6
L = 8
L = 10
ROC-AUC
100%
99%
98%
97%
96%
95%
94%
96.5
97.5
98.0
97.7
97.9
98.4
98.4
97.6
98.4
96.2
98.4
98.6
96.3
97.0
97.3
98.5
97.8
97.6
6 qubits
8 qubits
10 qubits
12 qubits
14 qubits
16 qubits
Zero-day detection (unseen attacks)
Classical baseline (88.8%)
L = 6
L = 8
L = 10
ROC-AUC
90%
89%
88%
87%
86%
85%
84%
86.7
85.9
87.8
86.4
87.1
88.3
88.2
87.5
88.2
86.1
88.2
88.2
88.9
87.1
88.9
88.8
89.1
88.2
6 qubits
8 qubits
10 qubits
12 qubits
14 qubits
16 qubits